

MAE HONG SON, Thailand

WMO: 483000

Lat: 19.301N

Lon: 97.976E

Elev: 929

StdP: 14.21

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.4	57.2	46.2	47.8	88.0	49.6	54.4	82.5	7.7	77.6	7.1	77.7	0.2	80	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	25.2	102.5	74.0	100.3	73.7	97.9	73.3	80.8	91.0	80.0	89.8	79.4	88.8	3.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.2	151.8	85.2	77.5	148.1	84.1	77.1	145.9	83.4	45.3	91.1	44.5	90.2	43.9	89.2	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	7.1	5.9	51.7	105.7	2.7	2.6	49.8	107.6	48.3	109.1	46.8	110.5	44.8	112.4		
(5)			50.4	83.1	2.5	1.3	48.6	84.0	47.1	84.8	45.7	85.5	43.8	86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	80.7	72.6	75.7	81.7	87.7	86.0	83.8	82.3	82.0	82.6	81.7	78.1	73.5		
	DBStd	5.38	2.79	2.92	3.35	3.51	3.88	2.43	2.19	1.96	1.86	2.08	3.13	3.70		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	1	0	0	0	0	0	0	0	0	0	0	0	1		
	CDD50	11190	701	718	983	1130	1117	1014	1001	993	977	984	843	729		
	CDD65	5717	237	298	518	680	652	564	536	528	527	519	393	265		
	CDH74	56937	1660	2910	6135	9147	7665	5874	5028	4821	4884	4476	2783	1554		
	CDH80	25473	538	1420	3597	5605	3986	2297	1631	1513	1723	1723	1007	433		
(14) Wind	WSAvg	1.3	1.2	1.2	1.4	1.8	1.6	1.5	1.4	1.3	1.3	1.3	1.1	1.1		
(15) Precipitation	PrecAvg	50.7	0.4	0.2	0.8	2.2	6.9	7.2	8.9	9.7	7.9	4.6	1.6	0.4		
	PrecMax	66.1	2.9	1.8	5.1	5.4	14.3	11.4	16.3	14.4	14.3	10.1	6.8	2.4		
	PrecMin	33.0	0.0	0.0	0.0	0.2	1.0	4.3	5.0	5.0	3.2	1.3	0.0	0.0		
	PrecStd	8.0	0.7	0.4	1.3	1.7	2.9	2.1	2.6	2.6	2.9	2.2	1.5	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	96.4	101.4	106.0	105.1	97.0	94.2	92.8	93.6	93.6	91.6	89.5		
		MCWB	69.8	68.5	70.8	74.8	76.0	78.3	78.9	79.0	78.7	77.8	75.5	72.5		
	2%	DB	87.8	93.7	99.4	103.8	101.2	94.5	91.6	90.7	91.7	91.8	89.7	86.5		
		MCWB	68.7	68.6	70.7	73.9	75.9	78.6	78.4	78.6	78.5	77.4	75.3	71.3		
	5%	DB	85.7	91.6	97.7	101.9	98.1	91.9	89.6	89.2	90.1	90.4	88.0	84.6		
		MCWB	68.2	68.0	70.4	73.8	76.8	78.2	78.1	78.2	78.2	77.2	74.5	70.6		
	10%	DB	83.2	89.2	95.5	99.4	95.1	89.8	87.7	87.4	88.3	88.7	86.1	82.4		
		MCWB	67.5	67.9	69.9	73.7	77.1	77.9	77.7	77.7	77.8	76.9	73.9	69.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.2	74.4	76.7	80.1	81.3	81.6	80.9	81.0	81.2	80.6	78.6	75.6		
		MCDB	82.8	88.3	95.2	97.0	95.2	92.0	89.7	89.3	89.8	90.0	88.0	84.8		
	2%	WB	72.1	72.4	74.9	78.4	80.0	80.3	79.9	79.9	80.1	79.4	77.2	74.2		
		MCDB	81.2	86.6	92.5	95.4	93.3	90.4	88.3	87.8	88.6	88.5	85.6	82.4		
	5%	WB	70.6	70.8	73.6	77.4	79.2	79.5	79.0	79.1	79.3	78.5	76.2	72.8		
		MCDB	80.0	85.0	90.2	94.1	91.9	88.7	86.9	86.4	87.1	87.3	84.1	80.2		
	10%	WB	69.2	69.5	72.4	76.4	78.3	78.7	78.3	78.3	78.5	77.7	75.2	71.5		
		MCDB	79.0	83.6	88.5	92.3	89.9	87.1	85.2	84.8	85.8	85.6	82.8	78.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.7	31.3	30.5	25.2	18.2	13.6	12.1	12.1	13.7	16.1	18.7	21.3		
		MCDBR	28.0	33.3	32.0	27.6	22.6	16.5	14.7	14.3	15.6	17.9	19.4	21.9		
		MCWBR	11.0	11.5	10.0	8.2	6.8	5.9	5.9	6.1	6.3	6.9	7.7	8.7		
	5% WB	MCDBR	23.3	29.5	28.1	24.0	18.9	14.4	12.9	13.0	14.2	16.2	17.4	19.1		
		MCWBR	10.1	11.1	9.7	8.0	6.8	6.0	6.0	6.1	6.4	6.9	7.5	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.398	0.471	0.667	0.656	0.518	0.452	0.438	0.450	0.469	0.491	0.420	0.401		
	taud		2.312	2.106	1.638	1.729	2.133	2.359	2.398	2.350	2.276	2.172	2.326	2.332		
	Ebn at Noon		278	262	217	222	252	267	271	270	264	253	269	273		
	Edh at Noon		38	50	82	76	50	40	38	40	43	46	38	37		
(44) All-Sky Solar Radiation	RadAvg		1514	1789	1860	1912	1695	1429	1301	1305	1431	1459	1483	1402		
	RadStd		67	57	108	113	138	95	120	76	56	75	104	87		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.29	N/A	N/A	+175	+176
(47) Regional (6 neighbors)	+0.52	+1.06	N/A	N/A	+0.27	+0.38	N/A	N/A	+265	+266

Nomenclature: See separate page